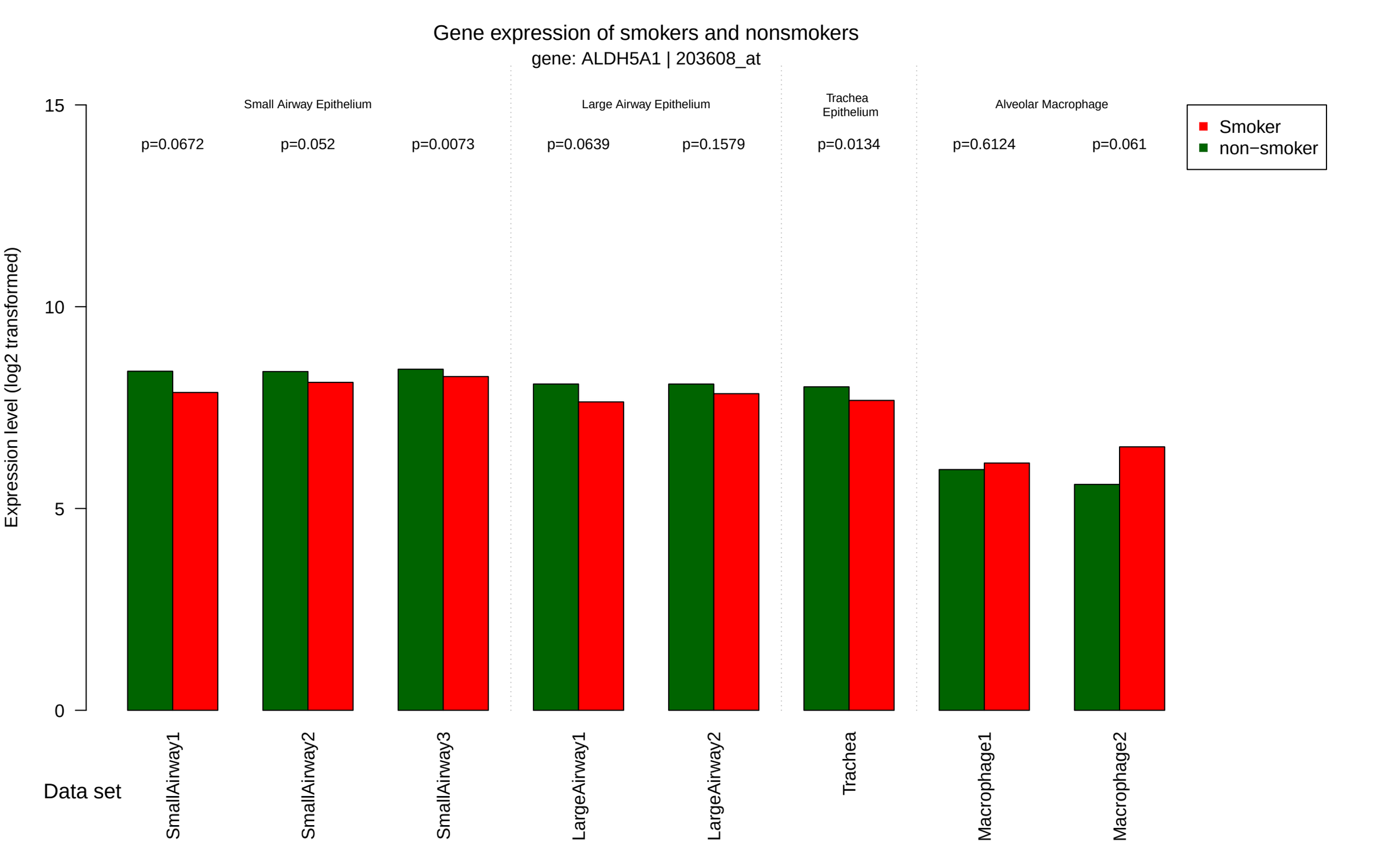
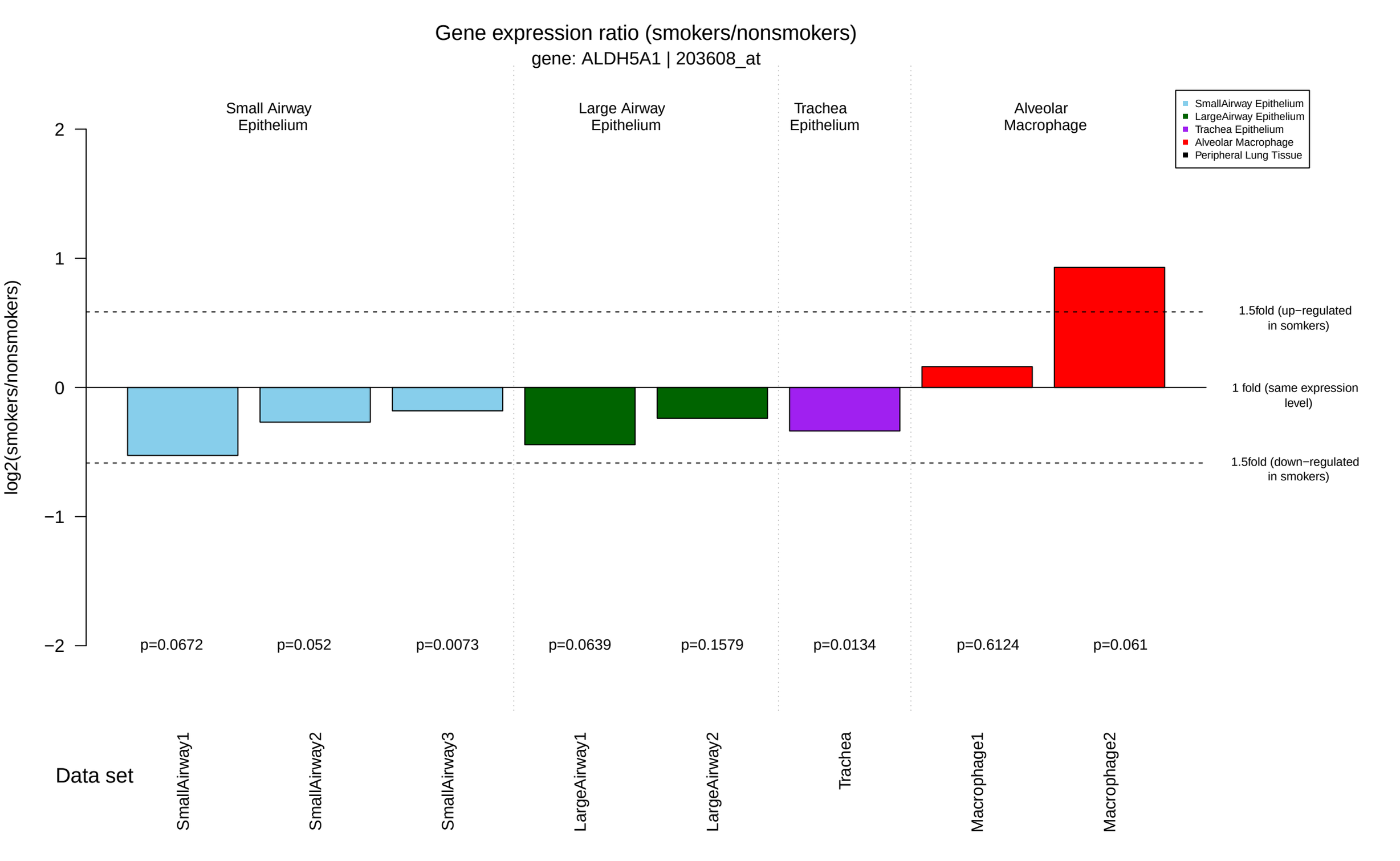
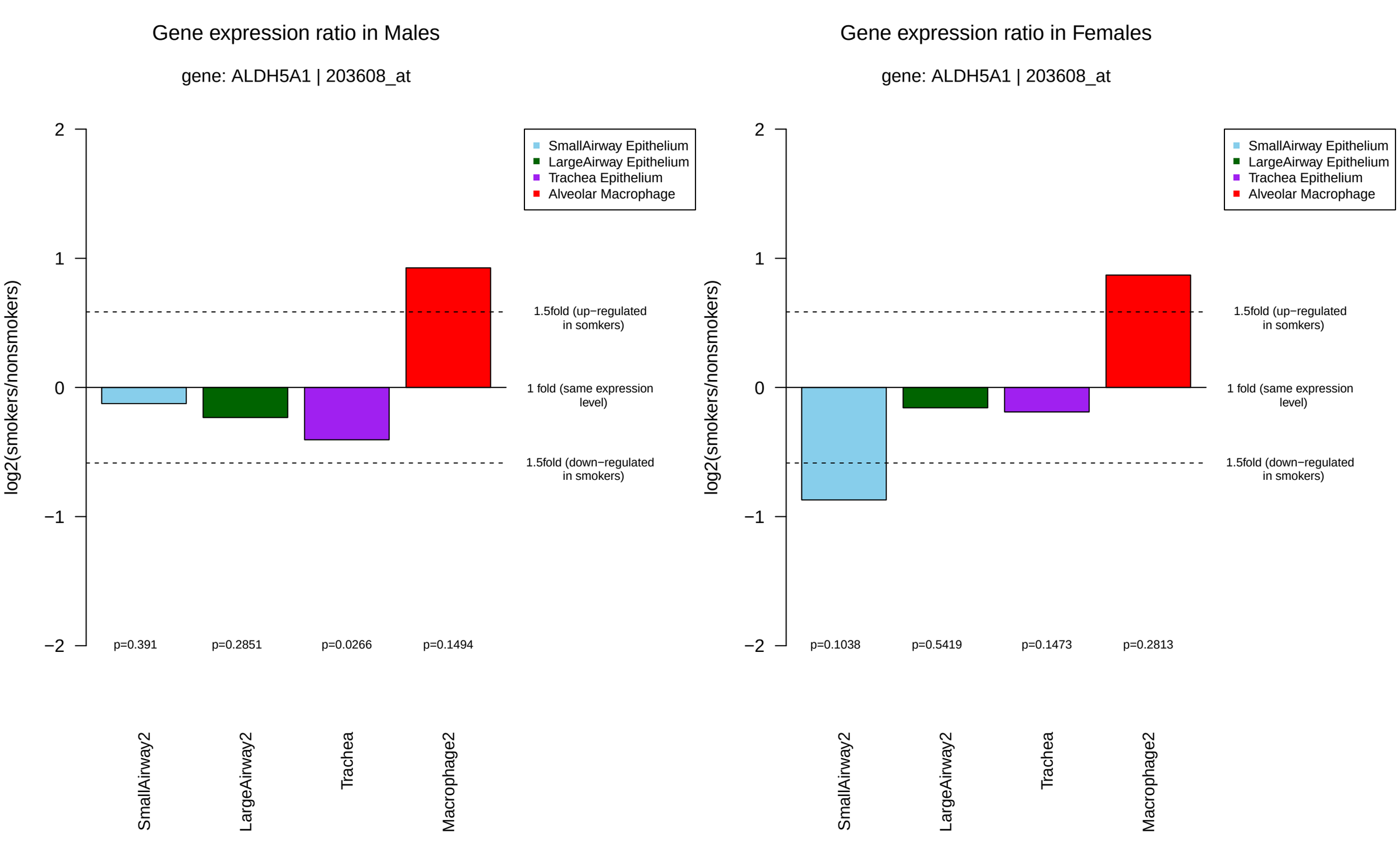
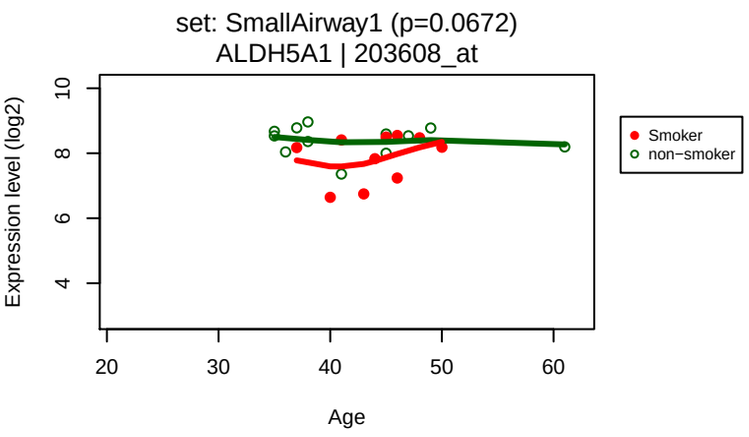
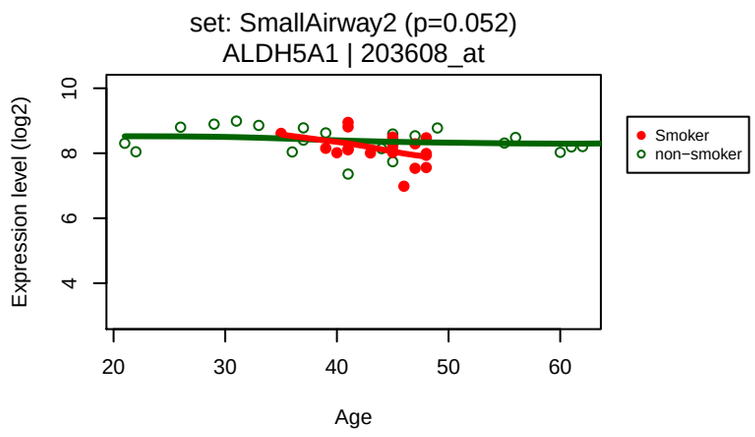
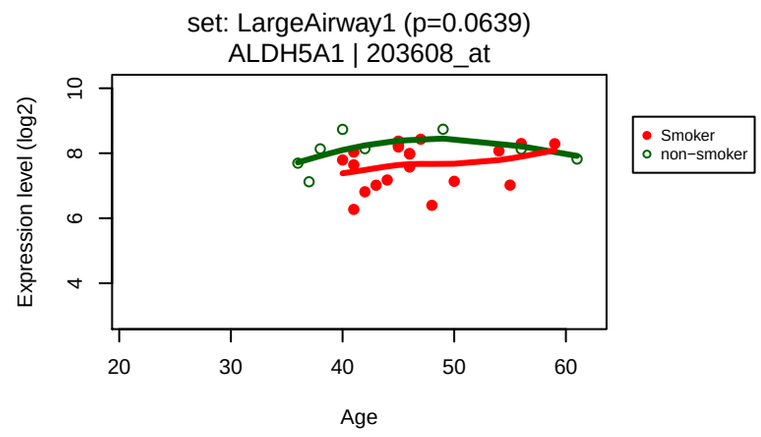

Gene expression of smokers and nonsmokers
gene: ALDH5A1 | 203608_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0672
p=0.052
p=0.0073
p=0.0639
p=0.1579
p=0.0134
p=0.6124
p=0.061
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

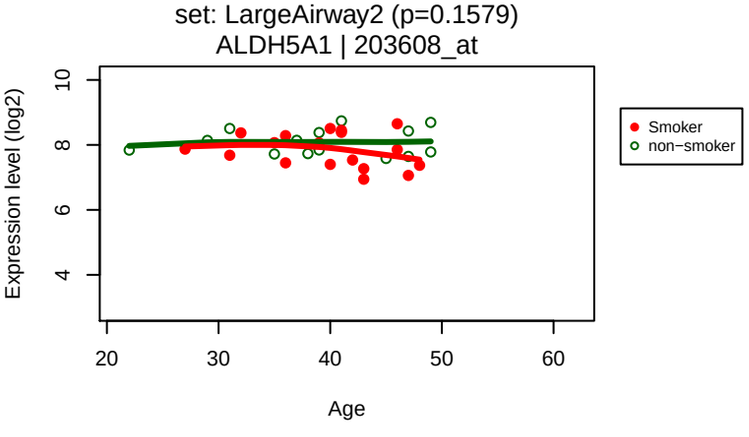

Gene expression ratio (smokers/nonsmokers)
gene: ALDH5A1 | 203608_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0672
p=0.052
p=0.0073
p=0.0639
p=0.1579
p=0.0134
p=0.6124
p=0.061
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

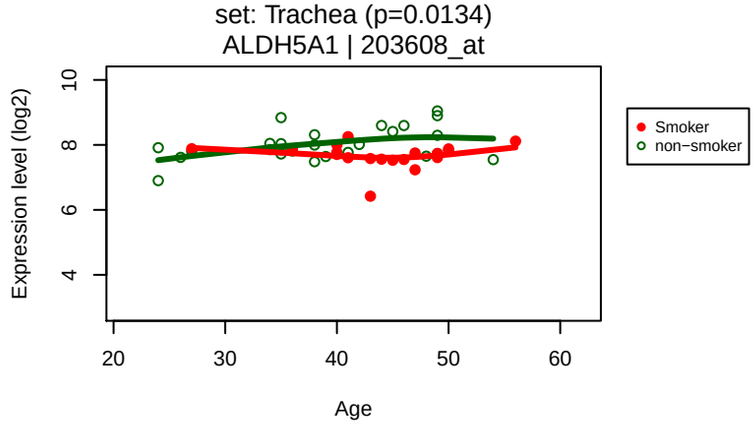

Gene expression ratio in Males
gene: ALDH5A1 | 203608_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.391
p=0.2851
p=0.0266
p=0.1494
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH5A1 | 203608_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1038
p=0.5419
p=0.1473
p=0.2813
SmallAirway2
LargeAirway2
Trachea
Macrophage2

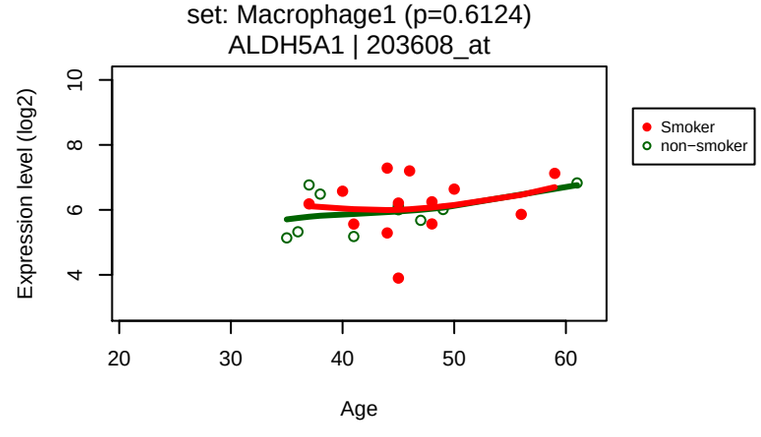

set: SmallAirway1 (p=0.0672)
ALDH5A1 | 203608_at
Expression level (log2)
Age
Smoker
non−smoker

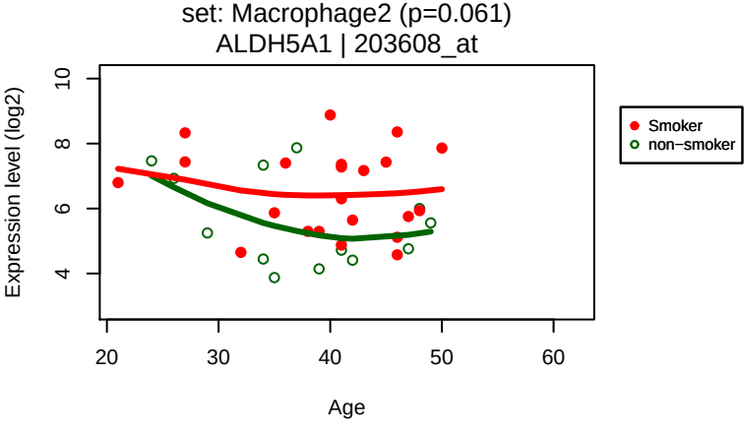

set: SmallAirway2 (p=0.052)
ALDH5A1 | 203608_at
Expression level (log2)
Age
Smoker
non−smoker

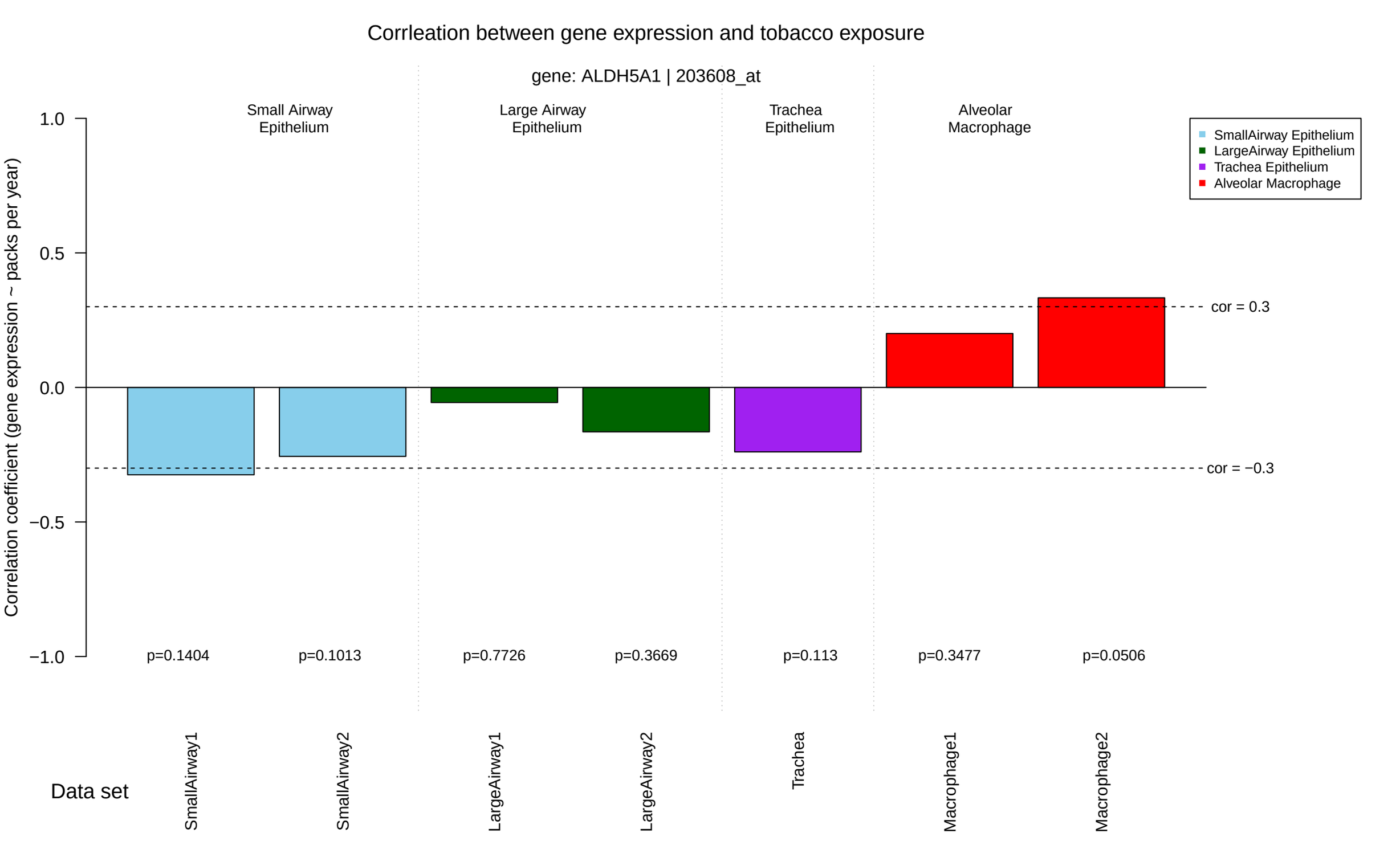

set: LargeAirway1 (p=0.0639)
ALDH5A1 | 203608_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1579)
ALDH5A1 | 203608_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0134)
ALDH5A1 | 203608_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6124)
ALDH5A1 | 203608_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.061)
ALDH5A1 | 203608_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH5A1 | 203608_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1404
p=0.1013
p=0.7726
p=0.3669
p=0.113
p=0.3477
p=0.0506
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2